🎬 Netflix Content Catalog Analysis

An end-to-end exploratory data analysis (EDA) of the Netflix global content catalog — **8,807 titles** spanning movies and TV shows released between 1925 and 2021 — built using a multi-tool analytics pipeline: **Python, SQL (PostgreSQL), and Power BI**.

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📌 Project Overview

This project analyzes Netflix's content strategy across four key lenses:

- **Content composition** — Movies vs. TV Shows
- **Geographic distribution** — where content is produced globally
- **Genre & rating trends** — audience maturity and genre popularity
- **Catalog growth over time** — release trends vs. platform addition trends

The goal is to simulate a real-world, business-ready analytics workflow — from raw data cleaning to stakeholder-facing dashboards — that supports decisions around content acquisition, regional investment, and catalog expansion.

📊 Key Findings

Metric	Value
Total Titles	8,807
Movies Share	69.6% (6,131 titles)
TV Shows Share	30.4% (2,676 titles)
Countries Represented	125+
Top Content Country	United States (3,689 titles)
Peak Release Year	2018 (1,147 titles)
Peak Addition Year	2019 (2,016 titles added)

Metric	Value
Most Common Rating	TV-MA (3,207 titles)
Top Genre	International Movies (2,752 titles)
Most Prolific Director	Rajiv Chilaka (22 titles)

Highlights:

- Movies outnumber TV shows nearly **2.3:1**, reflecting a historically film-first acquisition strategy.
- The **US and India** are the two dominant content markets.
- **South Korea (73.6%)** and **Japan (62.6%)** lean heavily toward TV-show content — strong drama/anime markets.
- Platform additions peaked in **2019** and have since declined, suggesting a pivot from volume growth to curation and originals.
- **72 countries** have fewer than 10 titles — potential white space for regional content investment.

🛠️ Tools & Tech Stack

Tool	Purpose
Python (Pandas, Matplotlib)	Data cleaning, EDA, and visualization across 29 business questions
PostgreSQL	Set-based aggregations, window functions (`RANK`, `LAG`), YoY growth analysis
Power BI	Interactive, stakeholder-facing dashboard with filtering and drill-down

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📁 Repository Structure

├── netflix_titles_0.xlsx # Source dataset (8,807 titles × 12 attributes)

├── netflix_Analysis_1.ipynb # EDA Q1–10: content mix, countries, genres, ratings, directors

├── Netflix_Analysis_2.ipynb # EDA Q11–20: trends, genre growth, country specialization

├── Netflix_Analysis_3.ipynb # EDA Q21–29 + PostgreSQL data load via SQLAlchemy

├── netflix_SQL.sql # 10 core SQL business questions incl. window functions

├── netflix_dashboard.pbix # Power BI interactive dashboard

├── Netflix_Content_Analysis_Report.pdf # Full written analysis report

├── Netflix-Content-Catalog-Analysis.pptx # Presentation summary of findings

└── README.md # Project documentation (this file)

🧹 Data Cleaning Steps

- Parsed the `duration` field — stripped `" min"` and converted to numeric minutes for movies.
- Exploded multi-value fields (`country`, `cast`, `director`, `listed_in`) from comma-separated strings into individual rows for accurate counting.
- Converted `date_added` to a proper datetime type.
- Standardized and trimmed whitespace from categorical fields (e.g., country names) prior to grouping.

🐍 Python Analysis (Jupyter Notebooks)

Split across three notebooks covering **29 structured business questions**:

1. **Notebook 1** — Content mix, top countries, top genres, rating distribution, top directors
2. **Notebook 2** — Time-based trends, decade-over-decade genre shifts, country content-type specialization
3. **Notebook 3** — Advanced questions + loading the cleaned dataset into PostgreSQL via SQLAlchemy

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🗄️ SQL Analysis (PostgreSQL)

`netflix_SQL.sql` contains 10 core business questions demonstrating relational/set-based analysis:

#	Business Question	SQL Technique
1	Top 10 directors by movie count	`GROUP BY`, `COUNT`, `ORDER BY`, `LIMIT`
2	Country ranking by title volume	`UNNEST` array split, `RANK() OVER()`
3	Longest movie in the catalog	`CAST` + `ORDER BY DESC`, `LIMIT 1`
4	Movies released after 2020	Filtering + `ORDER BY`
5	Directors active in both formats	`HAVING COUNT(DISTINCT type) = 2`
6	Title count by rating	`GROUP BY`, `COUNT`
7	Duplicate title detection	`GROUP BY` + `HAVING COUNT(*) > 1`
8	Average movie duration by country	`UNNEST`, `AVG`, `ROUND`
9	Top 5 genres	`UNNEST` array split, `GROUP BY`
10	Year-over-year growth (count & %)	`LAG() OVER()`, percentage calculation

Setup: Load `netflix_titles_0.xlsx` into a PostgreSQL table named `netflix_analysis`, then run the queries in `netflix_SQL.sql` sequentially.

📈 Power BI Dashboard

`netflix_dashboard.pbix` consolidates the key metrics — content mix, country distribution, genre popularity, and yearly growth — into an interactive, filterable dashboard for non-technical stakeholders to explore by year, country, or content type.

💡 Recommendations

- Continue investment in **Indian and South Korean** original productions given their strong representation and TV engagement.
- Evaluate the **72 underrepresented markets** (<10 titles each) for regional content-acquisition opportunities.
- Balance the movie-heavy catalog with further **TV-series investment**, especially in declining genres (Dramas, Documentaries).
- Use the PostgreSQL + Power BI layers as a **recurring reporting pipeline**, refreshed as new titles are added.

📄 Full Report

For the complete written analysis with all charts and figures, see **`Netflix_Content_Analysis_Report.pdf`**. A condensed visual summary is available in **`Netflix-Content-Catalog-Analysis.pptx`**.

👤 Author

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